

# Frontier AI for Actuaries.

AI helps actuaries perform complex, multi-step actuarial tasks with every step documented and reproducible.

## THE WORKBENCH

### One collaborative workspace for pricing, reserving, and filing

For all actuaries at admitted and specialty carriers, MGAs,  
and reinsurers.



# Actuarial work is not one workflow. It is a web of intricate, connected tasks.

The more steps an actuary delegates to AI, the less certain they can be of the output. Tesora enables full tracing, documentation, and audit, so the actuary remains in control.



**Philo Bishay** ✓ • 1st

Cofounder & Head of Product @ Tesora | Actuarial | Wharton MBA

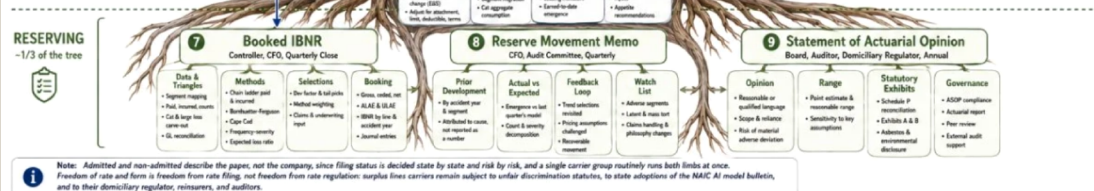
2W • ③

Every actuary dreads one question, "What do actuaries do?" I am happy to say we are no closer to an answer, but if you want to avoid further questions, send this:

## What Actuaries Deliver

Two questions. All decisions.  
What do we charge?  
What do we already owe?  
Actuaries own both.

**AT A GLANCE**  
10 Major Deliverables  
41 Branches  
163 Leaves  
121 Pricing Leaves (74%)  
42 Reserving Leaves (26%)



## Over 150 workflows are core to the actuarial function



■ 121 pricing ■ 42 reserving

one mark per workflow

### PRICING, A SAMPLE

Order of operations  
Base rate and factor structure  
Limit and deductible factors  
Schedule rating ranges  
IRPM guardrails  
Rating engine parity

### RESERVING, A SAMPLE

Chain ladder, paid and incurred  
Bornhuetter-Ferguson  
Cape Cod  
Development factor and tail picks  
Cat and large loss carve-out  
IBNR by line and accident year

## This is not a get-one-thing-right problem.

A rate filing or a booked reserve needs dozens of these correct, in sequence, each one feeding the next. Automate a single step and the chain still breaks at the handoff after it.



### One agentic harness across all of it

A system of agents keeps every task linked and every value cited, from ingest to filing. That orchestration is the hard part to build, and the harder part to replicate.

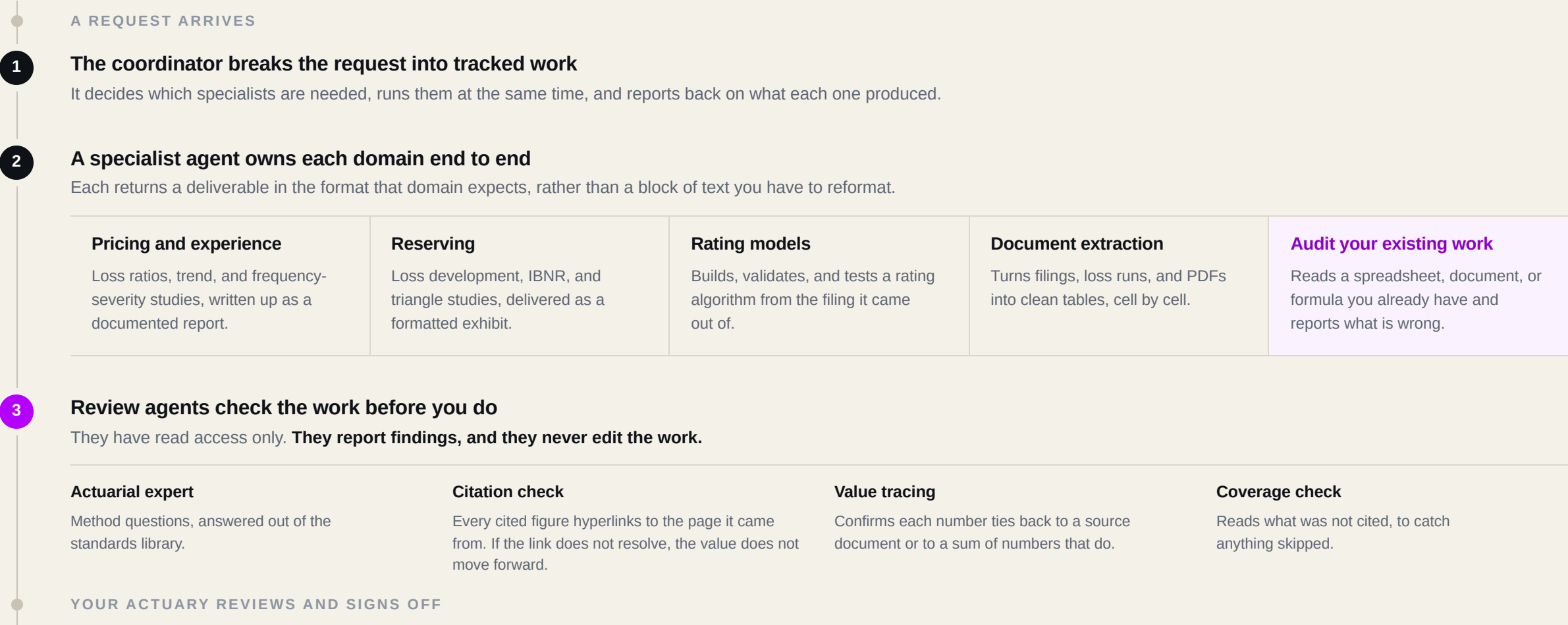


Federico Reyes Gómez and 675 others

23 comments • 28 reposts

# No single agent does the whole job, and none of it reaches you unchecked.

We watched how actuaries actually do the work, mapped it into sub-agents, wrote evals for each one, and put audit agents behind them so the model cannot cut a corner or invent a number.





# Rate changes in **days**, not months.

Source a rater from a SERFF filing, an Excel file, or a script. Tesora rebuilds it, tests it, and deploys it whenever underwriting is ready.

## BEFORE: BY HAND

# 4-6 months

From a requested change to a filed, live rater, and where the months actually go

- |    |                                                                                       |       |
|----|---------------------------------------------------------------------------------------|-------|
| 01 | <b>Spec the change</b><br>Actuary writes up the revised rates and factors             | DAYS  |
| 02 | <b>IT ticket, then the queue</b><br>The change waits for developer capacity           | WEEKS |
| 03 | <b>Re-code the rater</b><br>Engineering reimplements the logic by hand                | WEEKS |
| 04 | <b>Rebuild exhibits, 50 states</b><br>Each state reformatted by hand for SERFF        | WEEKS |
| 05 | <b>Manual QA</b><br>No test suite, so regressions are caught by eye                   | WEEKS |
| 06 | <b>Compliance sign-off</b><br>Actuarial and legal review the filing                   | DAYS  |
| 07 | <b>File and answer objections</b><br>Back and forth with each department of insurance | WEEKS |
| 08 | <b>Release window</b><br>Deploy only in the next scheduled window                     | WEEKS |

## AFTER: ON TESORA

# 1-2 days

Deploy as a GUI, a web interface, or an API.

**YOU GIVE** A plain-language change and the source rater, from SERFF, Excel, or a script.

**YOU GET** A tested rater with a working GUI, sample ratings, and a versioned API.

← Back to list Version 23 DEVELOPMENT

Inputs Computations Memo Review (23) Deployment

GUI Sample Ratings (15) Governance API

GUI template is valid

### Georgia Commercial Auto — Trucks/Tractors/Trailers

Liability + Physical Damage rating with zone-rated Long-Distance segment

**Vehicle & Classification**

Territory \* Size Class \*

101 Heavy Trucks

Business Use \* Radius \*

Commercial Local

Secondary Class \* Fleet \*

Not Otherwise Specified — Other No (Non-Fleet)

**Final Premium** **\$3,610.00**



# The triangle is squared before you sit down.

A quarterly close moves between actuarial, IT, and finance, and most of the calendar goes to assembling the data rather than judging it.

## BEFORE: BY HAND

~2 weeks

From close to a booked IBNR, and who is waiting on whom

### 01 Request the data

Actuarial files a ticket for the quarter's extract

IT

### 02 Wait on the extract

The pull sits behind other reporting work

DAYS

### 03 Reconcile to the ledger

Paid and incurred tied back to finance, by hand

FINANCE

### 04 Rebuild the triangles

Segment mapping and large-loss carve-outs redone each quarter

DAYS

### 05 Select factors and methods

Chain ladder, Bornhuetter-Ferguson, Cape Cod, tail

ACTUARIAL

### 06 Rebuild the exhibits

Roll-forward and movement memo reformatted by hand

DAYS

### 07 Peer review

A second actuary retraces the work from scratch

DAYS

### 08 Book and explain

Answer the CFO, the auditor, and the board

DAYS

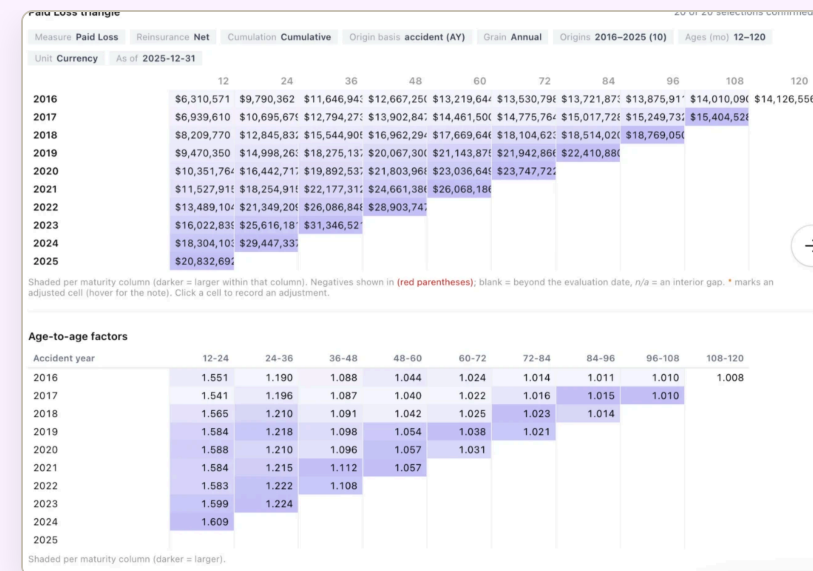
## AFTER: ON TESORA

3-4 hours

Triangles squared on ingest, so the quarter starts at the selection

**YOU GIVE** The loss run, as it comes out of the system.

**YOU GET** Reconciled triangles, method comparisons, and an IBNR with every value traced to source.



# Rebuild what a competitor charges, then **price your book through it.**

A filing gives you tables, rules, and inputs rather than a working model, and the order of operations is rarely spelled out. Turning that into something you can run your own risks through is the work.

## BEFORE: REBUILT BY HAND FROM SERFF

~1 week

From a competitor filing to a usable read, and where it goes wrong

- |    |                                                                                |         |
|----|--------------------------------------------------------------------------------|---------|
| 01 | <b>Pull the filings</b><br>An analyst searches SERFF, state by state           | HOURS   |
| 02 | <b>Read the rate manual</b><br>Rules, factors, and order of operations, by eye | DAYS    |
| 03 | <b>Rekey the rate tables</b><br>Hundreds of tables retyped into a spreadsheet  | DAYS    |
| 04 | <b>Rebuild the algorithm</b><br>The logic is inferred, then argued over        | DAYS    |
| 05 | <b>Price a sample</b><br>A dozen test risks, never the real book               | HOURS   |
| 06 | <b>Compare by hand</b><br>Deltas built in a one-off workbook nobody owns       | DAYS    |
| 07 | <b>Explain the gaps</b><br>Each difference traced back to a single factor      | DAYS    |
| 08 | <b>The read goes stale</b><br>They filed again while you were rebuilding       | ONGOING |

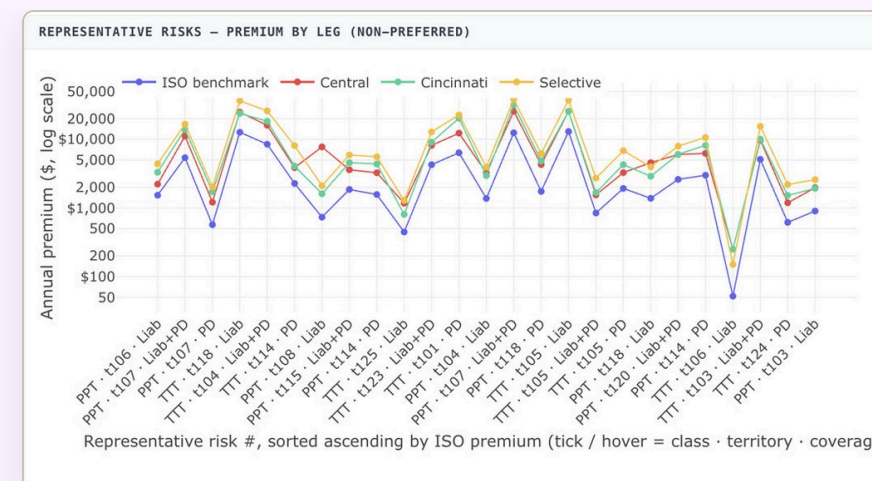
## AFTER: ON TESORA

10-15 minutes

An executable competitor rater, and your whole book priced through it

**YOU GIVE** A competitor, a state, and a line. Or the SERFF filing itself.

**YOU GET** Their rating logic running against your risks, every factor cited to the page it came from.



Built from public rate filings, and any additional data sources you provide.



# The full context is baked into the work.

Every session holds the data it used, the assumptions behind it, and the reasoning that got there. Reopen it years later and nothing has to be reconstructed, because the context was never separate from the answer.

The screenshot displays the Tesora platform interface. On the left is a sidebar with navigation options: Home, Work Tracking, ARTIFACTS (Folders, Reports, Raters, Routines), APPS (Compute St..., Underwriting, Rater Recon..., Competitive ..., GLM Studio, Triangle Bull..., All Apps, Support, Settings), and a user profile for 'vivek'. The main area is titled 'Bernegger 2024 — Integrated experience/exposure rating (ASTIN)' with a sub-header 'EX-VLT-bernegger-2024-integrated-rating'. Below this is a 'Rate Rule Form' table:

|                                 |            |
|---------------------------------|------------|
| Profit & Contingencies          | 7.0%       |
| Total expense & profit load     | 47.0%      |
| Expected Loss & LAE Ratio (ELR) | 53.0%      |
| Projected Ultimate Loss Ratio   | 54.7%      |
| Adequacy margin (ULR - ELR)     | 1.7% short |

Below the table, a 'Read' section states: 'the thin spot is the 47% load (30 pts of it commission), and the book runs -1.7 pts past its permissible loss ratio — a modest inadequacy with no cushion.' A 'Caveat' section follows: 'Caveat (carried from the exhibit): ILFs are relativities, not dollar prices; no competitor base rates or retention/deductible tables exist in the folder. Preliminary analyst work product, subject to credentialed-actuary review.' A note mentions 'The full CSV is downloaded at' and lists two files: 'CIC\_vs\_Peers\_ILF\_Competitive\_Benchmark.csv' and 'CIC vs Cyber Peers — ILF & Competitive Benchmark ...'. The 'CONTEXT' section at the bottom shows 'Bernegger 2024 — Integrated...' and a prompt 'Ask me anything...'. On the right, a document viewer shows a PDF titled 'bernegger-2024-integrated-experience-exposure-rating.pdf' with a 'Download' button. The document content includes a title 'Generic framework for a coherent integration of experience and exposure rating in reinsurance' by Stefan Bernegger, a summary, and an introduction section.

## Implementation

The models are wired into your loss runs, policy systems, and ledger, running inside your own environment.

## Governance

Who ran what, on which data, and who signed it. Finance and IT find what they need without waiting on an actuary to explain it.

## Compliance

Built to ASOP from the start, so a filing survives the objection, the examination, and the auditor years later.

## Continuity

When an actuary leaves, their assumptions, sources, and reasoning stay in the session. The next person picks up the thread.

Every request from the platform pings our chief actuary and CTO. Support is agents and people, so you are never left with a bot. A real person is available within two to three minutes on average, and we have had no unplanned downtime to date, which is the routing approach doing its job.



# What's the best way to **implement AI**?

Your implementation of AI today will dictate its value, translatability, and reproducibility going forward.

|                                  | <b>Actuarial consultants</b><br>Milliman, WTW, Oliver Wyman | <b>Your team with AI</b><br>In-house, on ChatGPT or Claude | <b>Actuarial software</b><br>hyperexponential, Earnix, Akur8 | <b>T Tesora</b><br>Frontier models, in your format   |
|----------------------------------|-------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------|
| <b>Turnaround</b>                | ✗ Weeks to months, in their queue                           | ≈ Quick drafts, still checked by hand                      | ✓ Fast and repeatable once it is live                        | ✓ Minutes to days, in your workflow                  |
| <b>Setup &amp; fit</b>           | ≈ A fresh engagement scoped each time                       | ✓ No one knows your book better                            | ✗ Heavy implementation on their platform                     | ✓ 60-day build in your VPC, on your own data         |
| <b>Your raters &amp; filings</b> | ✗ Generic deliverables in their template                    | ≈ Your format, rebuilt by hand each time                   | ≈ Rebuilt inside their system, their way                     | ✓ Your raters, versioned and ready to deploy         |
| <b>Source &amp; audit</b>        | ✓ Independent, credible workpapers                          | ✗ No source to trace, and the answer drifts                | ≈ Logged inside the platform                                 | ✓ Cell-level citations, append-only log              |
| <b>Scope</b>                     | ≈ Whatever you scope and pay for                            | ≈ One step at a time, gaps in between                      | ≈ Pricing or reserving, rarely filing too                    | ✓ Pricing, reserving, filing, and close in one place |



Strength



Partial or conditional



Gap

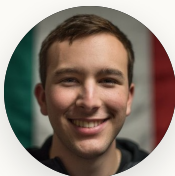


A team built at the intersection of frontier AI research and decades of actuarial experience, **the two disciplines this problem demands.**

**Vivek Rao**

CEO

Former private-equity investor and McKinsey consultant focused on P&C insurance.

**Federico Reyes**

CTO

Stanford AI researcher. Graph neural nets and relational data at Kumo AI, acquired by Nvidia.

**Philo Bishay**

CPO &amp; CHIEF ACTUARY

Former chief actuary at W. R. Berkley E&S. Capital modeler at Gen Re; built and sold a brokerage.

**Michael Palmich**

HEAD OF SALES

25+ years in the actuarial software space, most recently at Akur8 and Milliman.

**Eric Lam**

FOUNDING ENGINEER

MIT for CS, physics, and math, with **three actuarial exams** done.

## ADVISORY BOARD

**Ethan Genteman**

FCAS, MAAA

Chief Actuary, Palomar (NYSE: PLMR)

**Mario DiCaro**

FCAS, CERA

Head of Capital Modeling, Tokio Marine HCC

**Joanne Chen**

General Partner, Foundation Capital

BACKED BY



Leading AI investors, backing enterprise founders since 1995

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# Above all, we want to help you **win.**

We are a thought partner first. This technology is powerful, and it only helps once it is opinionated: built with a real point of view on how the work gets done and defended.

We are here to push the frontier forward with the right partners, not to sell you a license. A law firm does the legal work rather than selling software to lawyers, and we hold ourselves to that standard: **we take ownership of the outcome, not just the tooling.** Every deployment starts with a proof of concept anchored on your own success metrics, and we work with your team to find where the workbench sets the shared standard first.

Let's set up a call

[sales@tesora.ai](mailto:sales@tesora.ai)

## DE-RISKED BY DESIGN

- ✓ It runs in your own environment. We never hold your data.
- ✓ SOC 2 Type II, with the highest data-security controls.
- ✓ We clear your security review before any work touches a filing.

# Appendix



# Per actuary, the recurring month goes from 150 hours to about 60.

A full-time actuary has roughly 160 hours in a month, and most of them go to recurring work. Tesora compresses the mechanical grind, the data pulls, the reconciliations, the model updates, and hands back **upwards of 90 hours to every actuary**, to spend on judgment instead of assembly.

| Where one actuary's recurring month goes |                                   | <div><div></div> reclaimed</div> | <div><div></div> still hands-on</div> | ILLUSTRATIVE · MODELED |
|------------------------------------------|-----------------------------------|----------------------------------|---------------------------------------|------------------------|
| Data extraction & reconciliation         | <div><div></div><div></div></div> | 39                               | → 4                                   | 10×                    |
| Pricing model updates                    | <div><div></div><div></div></div> | 28                               | → 7                                   | 4×                     |
| Loss trend analysis                      | <div><div></div><div></div></div> | 23                               | → 12                                  | 2×                     |
| Reserve & loss development               | <div><div></div><div></div></div> | 21                               | → 12                                  | 1.7×                   |
| Management & board reporting             | <div><div></div><div></div></div> | 17                               | → 10                                  | 1.7×                   |
| Reinsurance program analysis             | <div><div></div><div></div></div> | 12                               | → 4                                   | 3×                     |
| Peer review & documentation              | <div><div></div><div></div></div> | 10                               | → 8                                   | 1.3×                   |
| Monthly total, per actuary               | <div><div></div><div></div></div> | 150                              | → ~60                                 | 2.5×                   |

## ~90 hours back per actuary, per month

A 2.5× blended throughput gain across the recurring month, with the same team and the same sign-off.

## Smarter, not harder

Time freed for strategy and selections rather than assembly.

## Unlock your actuaries

More products, more territories, and more actuarial capability from the same team.

Hours are per actuary and reflect a typical ~150 hour recurring workload within a roughly 160 hour month, with per-workflow gains from Tesora's engagement framework. Figures are modeled and confirmed against measured results during a build.



# We run on top of Claude, and we are model agnostic.

The model is the engine, not the product. What an actuary needs around it is infrastructure: the data wired in, the workflow decided, the cost controlled, and the record kept. That is the part we build, and we build it for you.

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## We guide the model

A raw model will answer anything, confidently. Ours is held to an actuarial method: which factors to consider, which checks to run, when to stop and ask. On a loss trend that means frequency and severity are fit separately and tested against your own history before anything is selected.

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## We wire the infrastructure

Your loss runs, policy systems, filings, and ledger are connected once and stay connected. A chat window starts empty every time. Ingest once here, and every study after it reads the same reconciled book.

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## The workflow is opinionated

Reserving and pricing have a right order of operations, and we encode it. Chain ladder, Bornhuetter-Ferguson, and Cape Cod run side by side, so the selection stays a choice your actuary makes rather than whatever the prompt happened to produce.

---

## We manage the cost

Running a real book through a frontier model naively is slow and expensive. Extraction runs on a small model and the judgment calls go to the strongest one, so the economics work at production scale.



We exist to build **your actuarial infrastructure**, not to serve the models or ourselves. When a better model ships, you get it. What stays is the system we built around your book.



## You buy the outcome. **We handle the models.**

Every firm is working out its AI spend internally right now. We run on the latest models, and our team of AI researchers and former investors knows both what these systems cost and how to get an outcome out of them for the least of it.



### **The most efficient model per outcome, at the right cost**

Each step is matched to the model that delivers it correctly for the least spend. Extraction runs small, judgment runs strong, and you are billed for the result rather than the tokens behind it.



### **Our incentives point the other way**

A model vendor is never going to tell you to buy less of what it sells. Because we are not paid by the token, your cost is not work we are merely free to do, it is work we have to solve for ourselves. **That is what makes us aligned over the long term.**



### **Always on the latest models, and always up**

Your workflows, data wiring, and audit trail live with us, so the strongest model arrives without a migration or a new vendor. A system built in-house on one provider stops when that provider does. We route across them, so **servicing an outage is our problem rather than your stopped quarter.**



### **We keep learning your systems**

The longer we work with your team, the more of your data, formats, and quirks the workbench encodes, and that knowledge carries across every upgrade.



### **Audits matter more as the models do more**

Better models take on more complex work, which raises rather than lowers the bar on proving it. Every value stays cited and reproducible, and **the advantage compounds for whoever moves first.**